

The k -digit number N is an Armstrong number if and only if the k -th power of each digit sums to N .

Given a positive integer N , return true if and only if it is an Armstrong number.

Example 1:

Input:

153

Output:

true

Explanation:

153 is a 3-digit number, and $153 = 1^3 + 5^3 + 3^3$.

Example 2:

Input:

123

Output:

false

Explanation:

123 is a 3-digit number, and $123 \neq 1^3 + 2^3 + 3^3 = 36$.

Example 3:

Input:

1634

Output:

true

Note:

$1 \leq N \leq 10^9$

```
1 #include <stdio.h>
2 #include <math.h>
3 int main(){
4     int n;
5     scanf("%d",&n);
6     int x=0,n2=n;
7     while(n2!=0){
8         x++;
9         n2=n2/10;
10    }
11    int sum=0;
12    int n3=n,n4;
13    while (n3!=0){
14        n4=n3%10;
15        sum=sum+pow(n4,x);
16        n3=n3/10;
17    }
18    if(n==sum){
19        printf("true");
20    }else{
21        printf("false");
22    }
23 }
```

	Input	Expected	Got	
✓	153	true	true	✓
✓	123	false	false	✓

Passed all tests! ✓

Question **2**

Correct

Marked out of
5.00

Flag question

Take a number, reverse it and add it to the original number until the obtained number is a palindrome. Constraints $1 \leq \text{num} \leq 999999999$ Sample Input 1 32 Sample Output 1 55 Sample Input 2 789 Sample Output 2 66066

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main(){
3     int rn,n,nt=0,i=0;
4     scanf("%d",&n);
5     do{
6         nt=n;
7         rn=0;
8         while(n!=0){
9             rn=rn*10+n%10;
10            n=n/10;
11        }
12        n=nt+rn;
13        i++;
14    }
15    while(rn!=nt || i==1);
16    printf("%d",rn);
17    return 0;
18 }
19
```

	Input	Expected	Got	
✓	32	55	55	✓
✓	789	66066	66066	✓

Passed all tests! ✓

Lemma 1. A connected graph G is a cycle if and only if it contains either 1 or 2 and 2 leaf nodes. When a graph has just one leaf node, then the leaf node is 1, and the root node is 2, and the graph is a cycle. When a graph has just two leaf nodes, then the leaf nodes are 1 and 2, and the root node is 3, and the graph is a cycle.

The program should accept a number *N* as input and display the odd less than *N* as output.



Here the lucky numbers are 1, 4, 11, 14, and the first lucky number is 11.


```

1 #include<stdio.h>
2 int main(){
3     int n=1,i=0,nt,co=0,e;
4     scanf("%d",&e);
5     while(i<e)
6     {
7         nt=n;
8         while(nt!=0)
9         {
10             co=0;
11             if(nt%10!=3 && nt%10!=4)
12             {
13                 co=1;
14                 break;
15             }
16             nt=nt/10;
17         }
18         if(co==0)
19         {
20             i++;
21         }
22         n++;
23     }
24     printf("%d",--n);
25     return 0;
26 }

```

	Input	Expected	Got	
✓	34	33344	33344	✓

Passed all tests! ✓